

Experiment No. 11

The Hall Effect

Objective

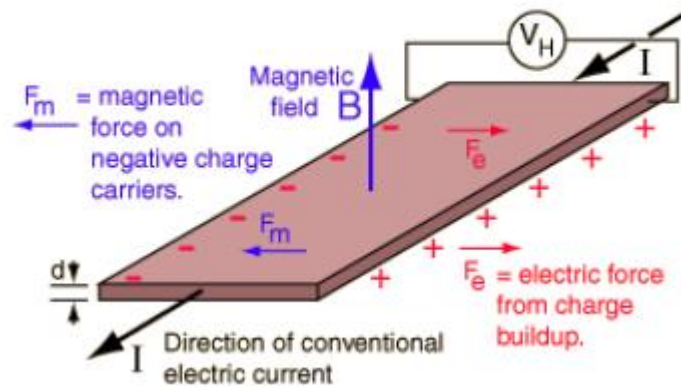
- To study the Hall Effect by measuring Hall voltage and current.
- Calculate the Hall coefficient R_H and the concentration of charge carriers (n).

Apparatus

- Power Supply for the electromagnet
- Constant Current Power Supply
- Semiconductor sample mounted on PCB.

Introduction

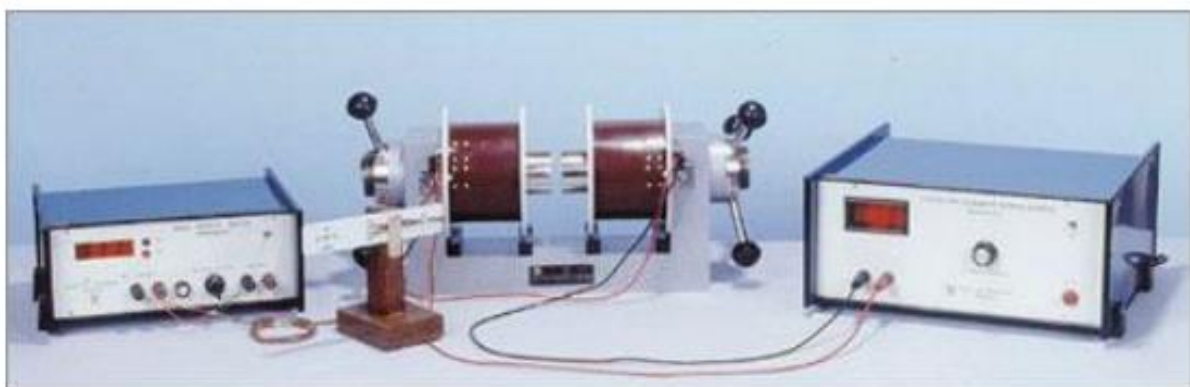
The Hall Effect is due to the nature of the current in a conductor. Current consists of the movement of many small charge carriers, typically electrons, holes, ions or all three. When a magnetic field is present, these charges experience a force, called the Lorentz force. When such a magnetic field is absent, the charges follow approximately straight, 'line of sight' paths between collisions with impurities, phonons, etc. However, when a magnetic field with a perpendicular component is applied, their paths between collisions are curved so that moving charges accumulate on one face of the material. This leaves equal and opposite charges exposed on the other face, where there is a scarcity of mobile charges. The result is an asymmetric distribution of charge density across the Hall element, arising from a force that is perpendicular to both the 'line of sight' path and the applied magnetic field. The separation of charge establishes an electric field that opposes the migration of further charge, so a steady electrical potential is established for as long as the charge is flowing. If an electric current flows through a conductor in a magnetic field, the magnetic field exerts a transverse force on the moving charge carriers which tends to push them to one side of the conductor. This is most evident in a thin flat conductor as illustrated.



A buildup of charge at the sides of the conductors will balance this magnetic influence, producing a measurable voltage between the two sides of the conductor. The presence of this measurable transverse voltage is called the Hall effect after E. H. Hall who discovered it in 1879. Note that the direction of the current I in the diagram is that of conventional current, so that the motion of electrons is in the opposite direction. This effect is very useful in determining:

- The nature of charge carries e.g. whether semiconductor is on n-type or p-type.
- Carrier concentration or the number density of charge carries.
- Mobility of charge carriers.

Experimental Setup



Procedure

1. Connect one pair of contacts of specimen on the opposite faces to the current source and other pair to the multimeter.
2. Switch on the power supply of electromagnet and calculate the magnetic flux density at the centre between the pole faces.
3. Place the specimen at the centre between the pole faces such that the magnetic field is perpendicular to the strip.
4. Pass the current (mA) from the current source through the specimen and measure the resulting hall voltage in the multimeter.
5. Increase the current through the specimen gradually and measure the corresponding Hall voltages.
6. The entire process can be repeated for different values of magnetic flux density.

Activity

Change the input current in electromagnets and note down the corresponding changes in hall voltage. Also Calculate the Hall coefficient and charge concentration using the formulas given below

B = Magnetic flex density in Tesla B= _____	
INPUT Hall Current (mA)	OUTPUT Hall Voltage (V)

i) Hall Coefficient:

$$R_H = \left(\frac{V_H}{I} \right) \frac{t}{B} =$$

Where,

V_H = Hall Voltage in volts

I = Current through the specimen in mA

t = thickness of specimen in meters ($t = 0.5\text{mm}$)

B = Magnetic flux density in Tesla

ii) Carrier Concentration:

$$n = \frac{1}{qR_H} =$$

Where,

q = Charge of electrons/holes

R_H = Hall Coefficient

Lab Exercise and Summary

Summary should cover Introduction, Procedure, Data Analysis and Evaluation.

[illegible]



Date: _____

LABORATORY SKILLS ASSESMENT (Psychomotor)

Total Marks: 100

Criteria (Max Marks)	Level 1 0% ≤ S < 50%	Level 2 50% ≤ S < 70%	Level 3 70% ≤ S < 90%	Level 4 90% ≤ S ≤ 100%	Score (S)
Procedural Awareness (20)	Selects inappropriate skills and/or strategies required by the task	Selects and applies appropriate skills and/or strategies required by the task with some errors	Selects and applies the appropriate strategies and/or skills specific to the task without significant errors	Selects and applies appropriate strategies and/or skills specific to the task without any error	
Practical Implementation (30)	Makes several critical errors in applying procedural knowledge of Measurement of Hall Voltage	Makes few critical errors in applying procedural knowledge of Measurement of Hall Voltage	Makes some non-critical errors in applying procedural knowledge of Measurement of Hall Voltage	Applies the procedural knowledge of Measurement of Hall Voltage in perfect ways	
Safety (10)	Requires constant reminders to follow safety procedures	Requires some reminders to follow safety procedures	Follows safety procedures with only minimal reminders	Routinely follows safety procedures	
Use of Tool/Equipment (20)	Uses tools, equipment and materials with limited competence	Uses tools, equipment and materials with some competence	Uses tools, equipment and materials with considerable competence	Uses tools, equipment and materials with a high degree of competence	
Participation to Achieve Group Goals (10)	Shows little commitment to group goals and fails to perform assigned roles	Demonstrates commitment to group goals, but has difficulty performing assigned roles	Demonstrates commitment to group goals and carries out assigned roles effectively	Actively helps to identify group goals and works effectively to meet them in all roles assumed	
Interpersonal Skills in Group Work (10)	Rarely interacts positively within a group, even with prompting	Interacts with other group members if prompted	Interacts with all group members spontaneously	Interacts positively with all group members and encourages such interaction in others	
Marks Obtained					

Instructor's Signature: _____

Date: _____

LABORATORY SKILLS ASSESMENT (Affective)

Total Marks: 40

Criteria (Max. Marks)	Level 1 $0\% \leq S < 50\%$	Level 2 $50\% \leq S < 70\%$	Level 3 $70\% \leq S < 90\%$	Level 4 $90\% \leq S \leq 100\%$	Score (S)
Introduction (5)	Very little background information provided or information is incorrect	Introduction is brief with some minor mistakes	Introduction is nearly complete, missing some minor points	Introduction complete and well-written; provides all necessary background principles for the experiment	
Procedure (5)	Many stages of the procedure are not entered on the lab report.	Many stages of the procedure are entered on the lab report.	The procedure could be more efficiently designed but most stages of the procedure are entered on the lab report.	The procedure is well designed and all stages of the procedure are entered on the lab report.	
Data Record (10)	Data is brief and missing significant pieces of information.	Data provides some significant information and has few critical mistakes.	Data is almost complete but has some minor mistakes.	Data is complete and relevant. Tables with units are provided. Graphs are labeled. All questions are answered correctly.	
Data Analysis (10)	Data is presented in very unclear manner. Error analysis is not included.	Data is presented in ways (charts, tables, graphs) that are not clear enough. Error analysis is included.	Data is presented in ways (charts, tables, graphs) that can be understood and interpreted. Error analysis is included.	Data are presented in ways (charts, tables, graphs) that best facilitate understanding and interpretation. Error analysis is included.	
Report Quality (10)	Report contains many errors.	Report is somewhat organized with some spelling or grammatical errors.	Report is well organized and cohesive but contains some grammatical errors.	Report is well organized and cohesive and contains no grammatical errors. Presentation seems polished.	
Marks Obtained					

LABORATORY SKILLS ASSESMENT (Cognitive)

Total Marks: 10

(If any)	
Marks Obtained	

Instructor's Signature: _____

Date: _____